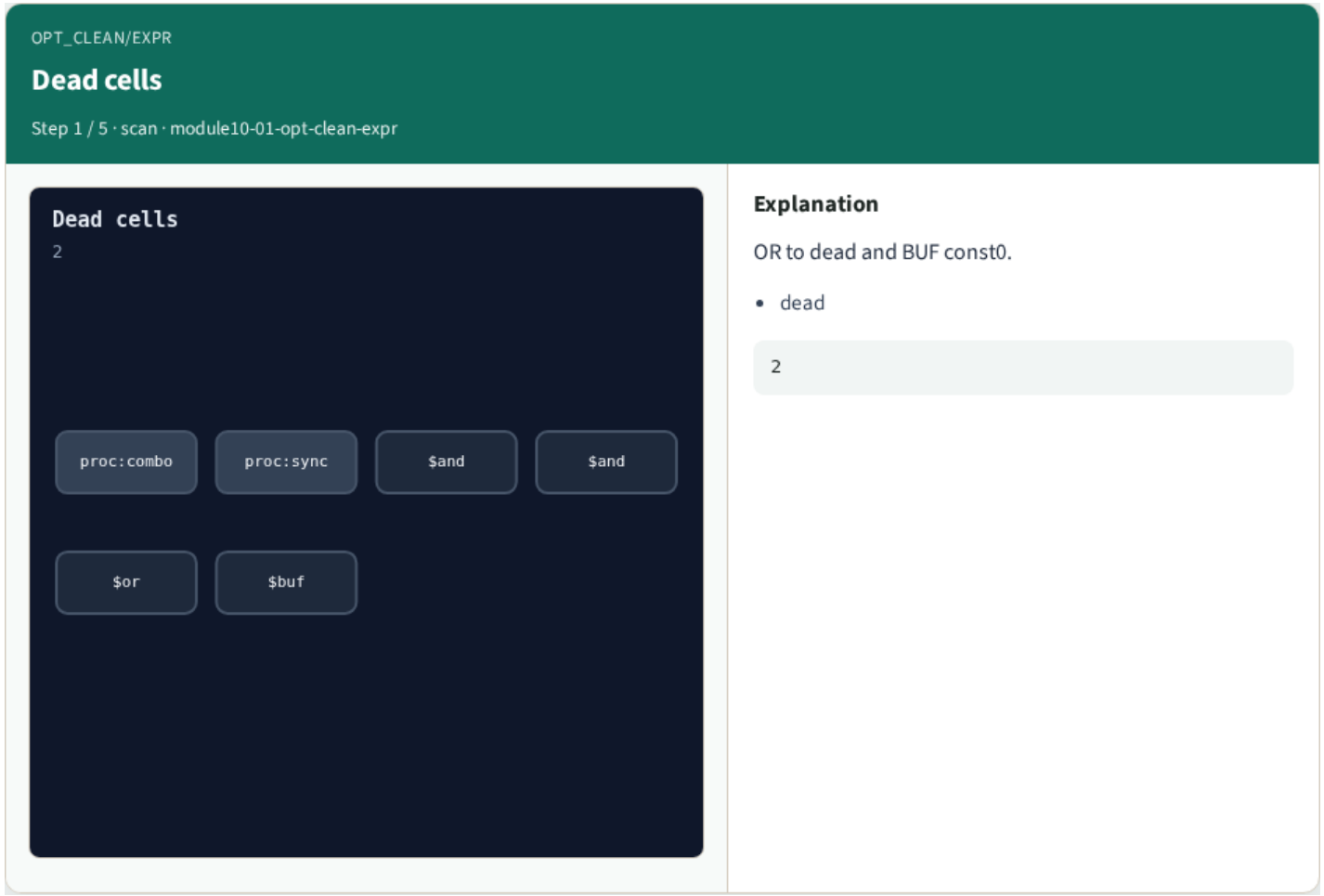
Why opt clean

Dead logic and constant buffers waste cells; clean and fold shrink the netlist

The idea

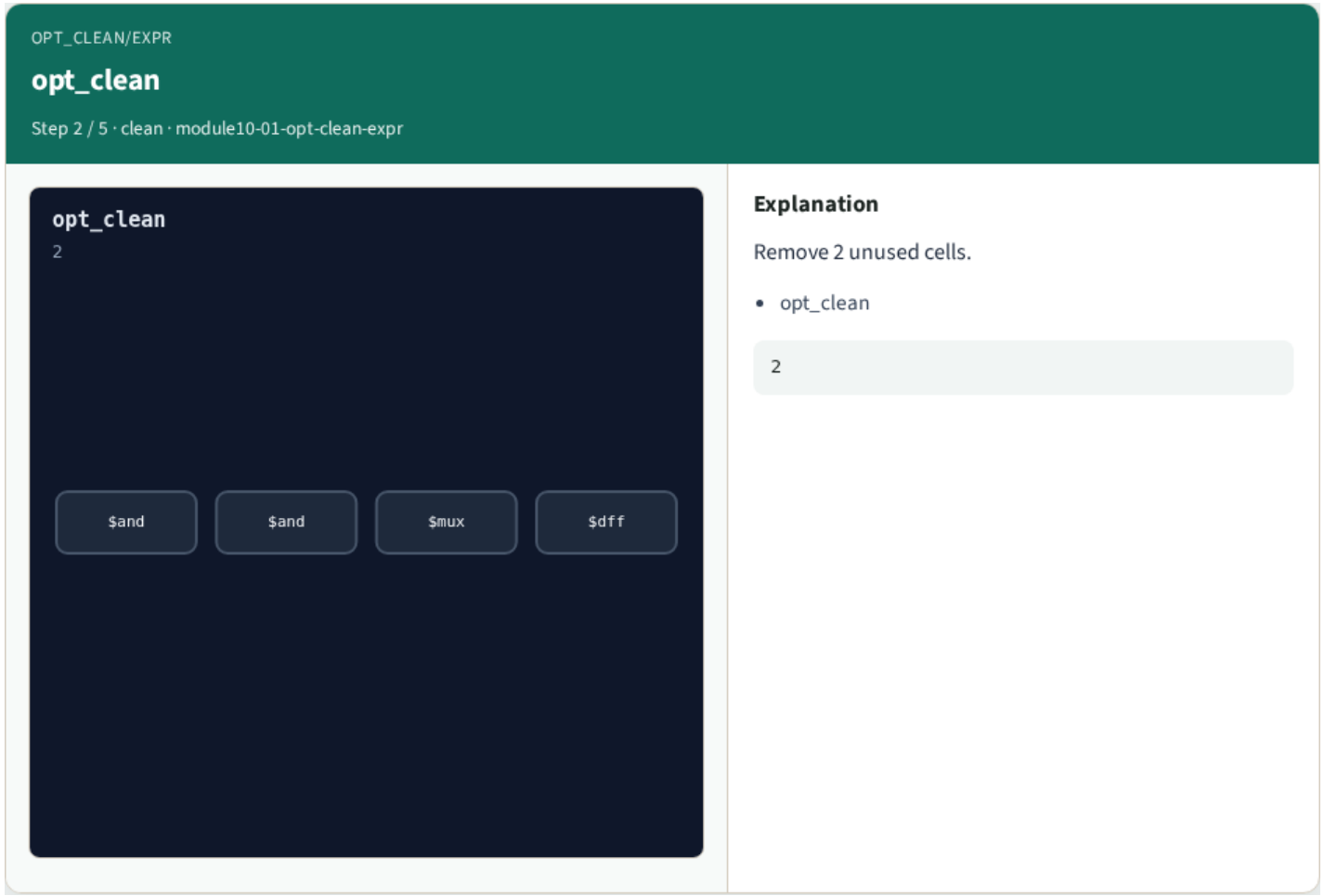
- GOLDEN: opt_clean removed=2; opt_expr folded=1

Dead cells



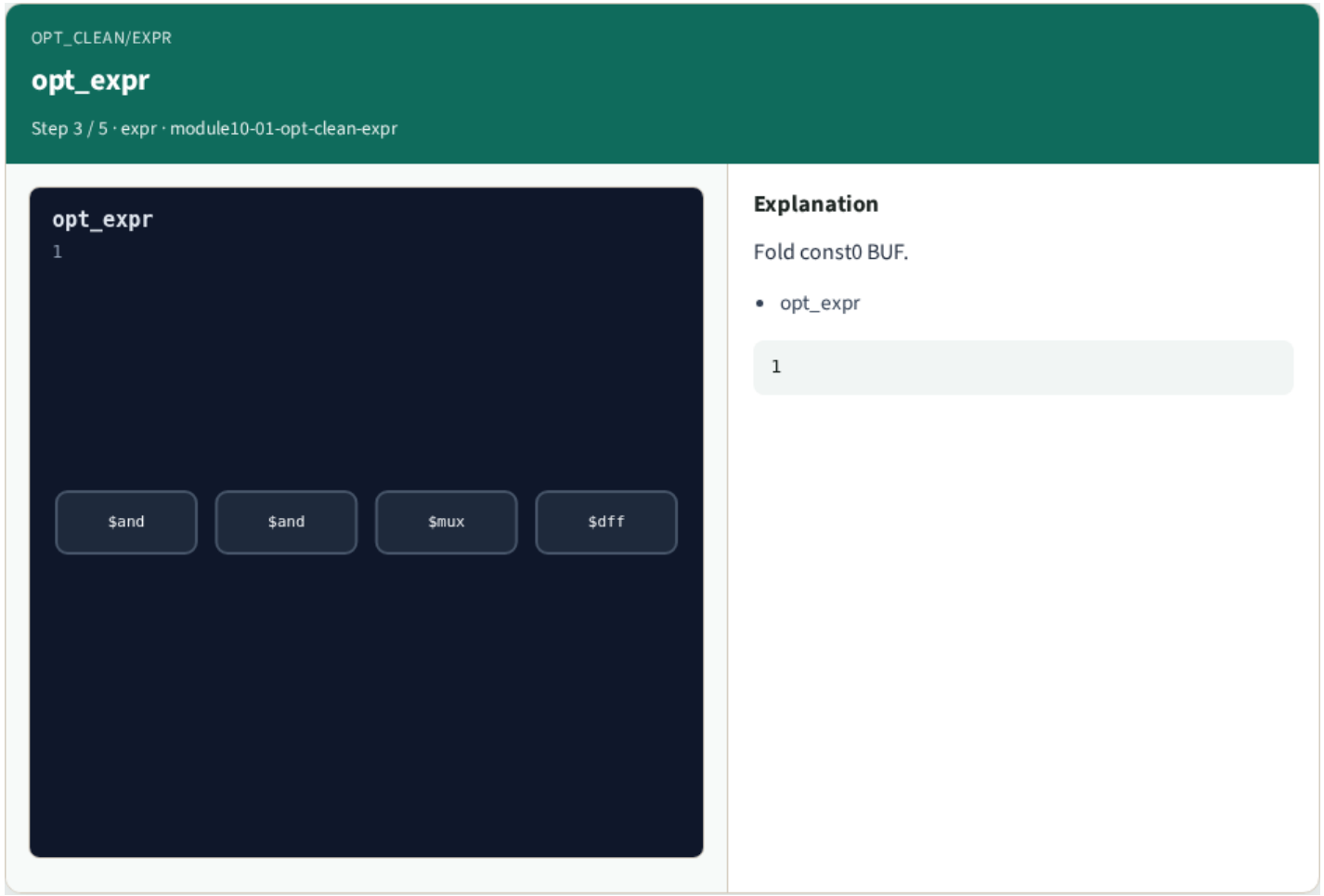
Dead cells

opt_clean



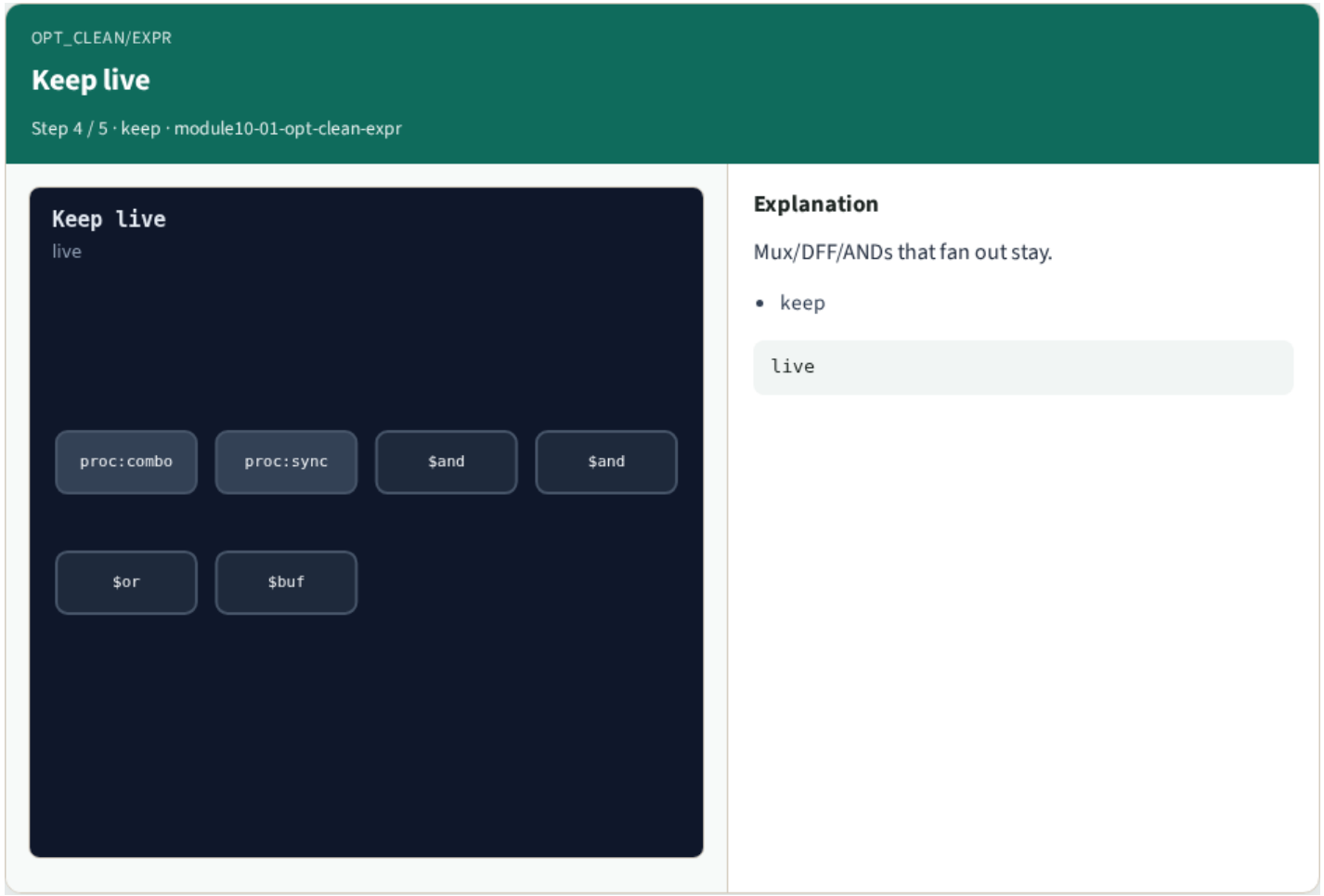
opt_clean

opt_expr



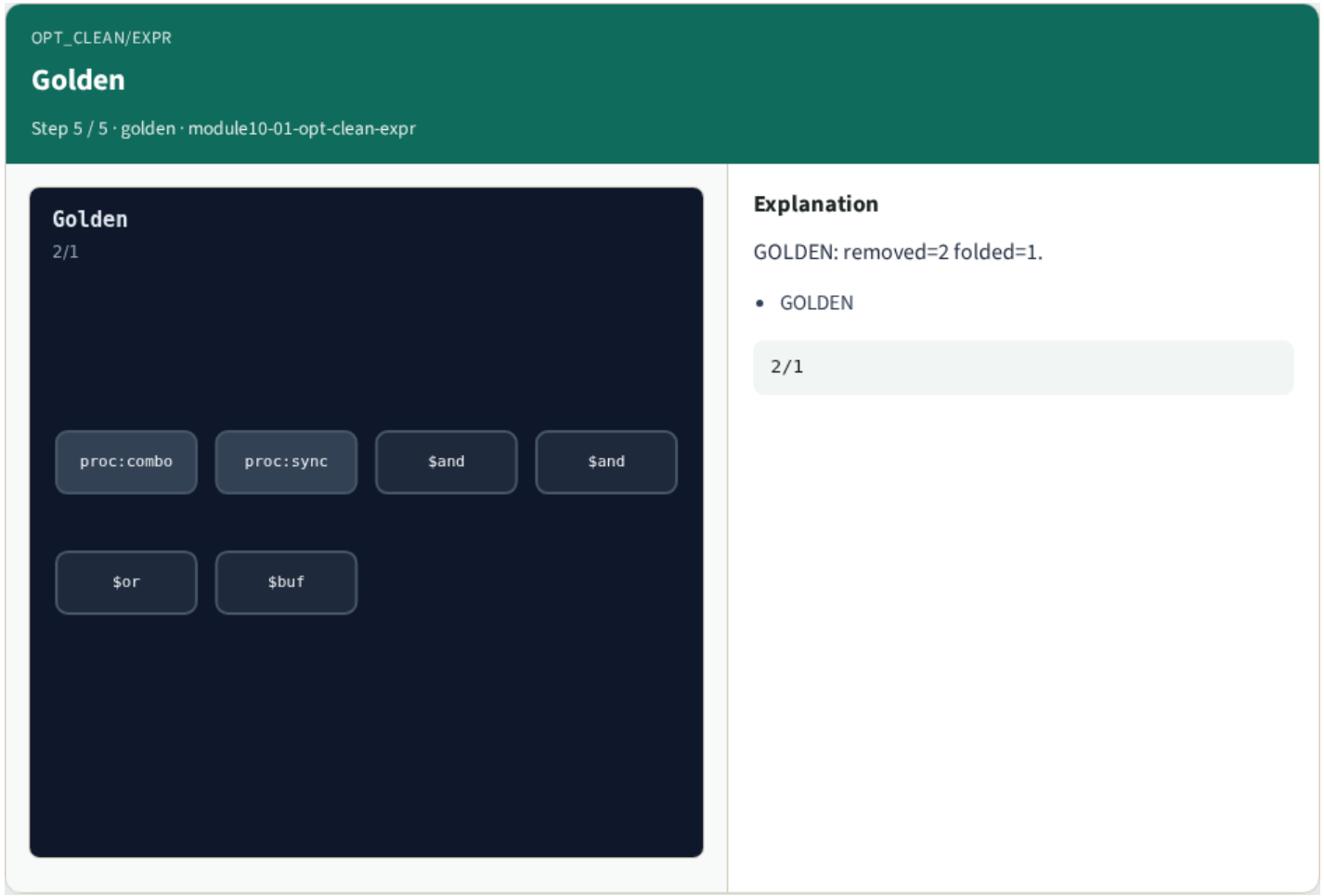
opt_expr

Keep live



Keep live

Golden



Golden

Browser lab track

- Open **opt-clean-expr**
- Remove dead cells and fold the const0 buffer

Implement track

- Call algo_opt_clean(); assert removed_count == 2
- Call algo_opt_expr(); assert folded_count == 1

Pitfalls

- Deleting live fanout
- Counting fold after clean already dropped the buffer

Try these

- From the common folder, run the algo solvers self-test, or call `algo_opt_clean` directly

Try these — try these

```
cd courses/learn_yosys/common
python3 algo_solvers.py
```

Pseudocode

- Here is the check in plain steps

Pseudocode — try these

```
mark used nets from POs backward
drop cells whose Y/Q never used (keep mux/dff)
fold dollar buf tied to const0
ok if removed==2 and folded==1
```

Next

- Complete the quiz for this part
- Next: opt_merge of duplicate ANDs